

Table 1: Selected Hyperparameters for Lapse Prediction Models

Model	Selected hyperparameters
Logistic Regression	$C = 1.2578$ $l1_ratio = 1.0000$ solver = saga max_iter = 1000
Decision Tree	max_depth = 9 min_samples_split = 129 min_samples_leaf = 36 criterion = gini
Random Forest	n_estimators = 396 max_depth = 8 min_samples_split = 174 min_samples_leaf = 72 max_features = sqrt subsample = 0.8771 criterion = entropy
XGBoost	n_estimators = 598 learning_rate = 0.1388 max_depth = 9 min_child_weight = 20 subsample = 0.6795 colsample_bynode = 0.8731 reg_alpha (L1) = 0.5394 reg_lambda (L2) = 0.1247
LightGBM (Full)	n_estimators = 598 learning_rate = 0.0818 num_leaves = 57 max_depth = 10 min_child_samples = 58 subsample = 0.8513 colsample_bynode = 0.5055 reg_alpha (L1) = 0.0010 reg_lambda (L2) = 0.0014
LightGBM (98%)	n_estimators = 499 learning_rate = 0.0669 num_leaves = 32 max_depth = 7 min_child_samples = 98 subsample = 0.8108 colsample_bytree = 0.5044 reg_alpha (L1) = 0.3526 reg_lambda (L2) = 0.3953
LightGBM (90%)	n_estimators = 300 learning_rate = 0.0669 num_leaves = 32 max_depth = 7 min_child_samples = 98 subsample = 0.8108 colsample_bytree = 0.5044 reg_alpha (L1) = 0.3526 reg_lambda (L2) = 0.3953
LightGBM (80%)	n_estimators = 300 learning_rate = 0.0669 num_leaves = 32 max_depth = 7 min_child_samples = 98 subsample = 0.8108 colsample_bytree = 0.5044 reg_alpha (L1) = 0.3526 reg_lambda (L2) = 0.3953
Multi-Layer Perceptron (MLP)	hidden_layer_sizes = 192 activation = tanh alpha (L2) = 0.0005 learning_rate_init = 0.0027 batch_size = 256 1